

Commerce XI Maths App

Theory (Pages 31–33)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 2 - Measures of Dispersion | Total questions: 32

Question 1 - Theory P31

Combined data means data obtained by

Options

- A. joining two or more data sets
- B. removing all frequencies
- C. using only quartiles
- D. drawing an ogive

Correct answer: joining two or more data sets

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives joining two or more data sets.

Question 2 - Theory P31

For samples of sizes n_1 and n_2 , combined size is

Options

- A. $n_1 + n_2$
- B. $n_1 \times n_2$
- C. $n_1 - n_2$
- D. $(n_1 + n_2)/(2)$

Correct answer: $n_1 + n_2$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $n_1 + n_2$.

Question 3 - Theory P31

The combined mean is

Options

- A. $(n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$
- B. $(\bar{x}_1 + \bar{x}_2)/(2)$ always
- C. $n_1\bar{x}_1 - n_2\bar{x}_2$
- D. $\sigma_1 + \sigma_2$

Correct answer: $(n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $(n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1 + n_2)$.

Question 4 - Theory P31

In the combined S.D. formula, d_1 equals

Options

A. $\bar{x}_1 - \bar{x}_c$

B. $\bar{x}_c - \bar{x}_2$

C. $\sigma_1 - \sigma_c$

D. $n_1 - n_2$

Correct answer: $\bar{x}_1 - \bar{x}_c$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\bar{x}_1 - \bar{x}_c$.

Question 5 - Theory P31

In the combined S.D. formula, d_2 equals

Options

A. $\bar{x}_2 - \bar{x}_c$

B. $\bar{x}_c - \bar{x}_1$

C. $\sigma_2 - \sigma_c$

D. $n_2 - n_1$

Correct answer: $\bar{x}_2 - \bar{x}_c$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $\bar{x}_2 - \bar{x}_c$.

Question 6 - Theory P31

Combined variance includes both within-group variance and

Options

A. differences between group and combined means

B. class limits only

C. quartiles only

D. ranges only

Correct answer: differences between group and combined means

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives differences between group and combined means.

Question 7 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45, the combined mean is

Options

- A. 38
- B. 34.5
- C. 35
- D. 69

Correct answer: 38

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 38.

Question 8 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45 and combined mean 38, d_1 is

Options

- A. -14
- B. 14
- C. -7
- D. 21

Correct answer: -14

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives -14.

Question 9 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45 and combined mean 38, d_2 is

Options

- A. 7
- B. -7
- C. 14
- D. 21

Correct answer: 7

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 7.

Question 10 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45 and standard deviations 6 and 11, the combined variance is approximately

Options

- A. 190.67
- B. 13.8
- C. 121
- D. 36

Correct answer: 190.67

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 190.67.

Question 11 - Solved Example P31

For groups of sizes 10 and 20 with means 24 and 45 and standard deviations 6 and 11, the combined standard deviation is approximately

Options

- A. 13.8
- B. 190.67
- C. 17
- D. 8.5

Correct answer: 13.8

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 13.8.

Question 12 - Solved Example P31-32

If the first group has 100 items and the combined group has 250, the second group contains

Options

- A. 150 items
- B. 100 items
- C. 250 items
- D. 350 items

Correct answer: 150 items

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 150 items.

Question 13 - Solved Example P31-32

If the group means are 45 and unknown and the combined mean is 51, the second mean is

Options

- A. 55
- B. 51
- C. 45
- D. 60

Correct answer: 55

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 55.

Question 14 - Solved Example P31-32

If the combined variance is 130, the second-group variance is

Options

- A. 144
- B. 12
- C. 130
- D. 49

Correct answer: 144

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 144.

Question 15 - Solved Example P31-32

The second-group standard deviation is

Options

- A. 12
- B. 144
- C. 11
- D. 14

Correct answer: 12

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 12.

Question 16 - Theory P32

Standard deviation depends on the

Options

- A. unit of measurement
- B. order of observations only
- C. name of the variable
- D. sample label

Correct answer: unit of measurement

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives unit of measurement.

Question 17 - Theory P32

Coefficient of variation is independent of the

Options

- A. unit of measurement
- B. mean
- C. standard deviation
- D. sample size

Correct answer: unit of measurement

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives unit of measurement.

Question 18 - Theory P32

Coefficient of variation is calculated as

Options

- A. $100 \times (\sigma)/(\bar{x})$
- B. $100 \times (\bar{x})/(\sigma)$
- C. $(\sigma^2)/(\bar{x})$
- D. $\bar{x} - \sigma$

Correct answer: $100 \times (\sigma)/(\bar{x})$

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives $100 \times (\sigma)/(\bar{x})$.

Question 19 - Theory P32

Coefficient of variation is expressed as a

Options

- A. percentage
- B. class interval
- C. frequency
- D. raw score

Correct answer: percentage

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives percentage.

Question 20 - Theory P32

A distribution with the smaller C.V. is more

Options

- A. consistent
- B. variable
- C. skewed
- D. dispersed

Correct answer: consistent

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives consistent.

Question 21 - Theory P32

A distribution with the larger C.V. is more

Options

- A. variable
- B. consistent
- C. compact
- D. homogeneous

Correct answer: variable

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives variable.

Question 22 - Theory P32

C.V. is particularly useful for comparing data measured in

Options

- A. different units or with different means
- B. the same order only
- C. equal frequencies only
- D. one class only

Correct answer: different units or with different means

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives different units or with different means.

Question 23 - Solved Example P32

Varad has mean 50 and S.D. 11. His C.V. is

Options

- A. 22%
- B. 11%
- C. 25%
- D. 50%

Correct answer: 22%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 22%.

Question 24 - Solved Example P32

Viraj has mean 58 and S.D. 16. His C.V. is approximately

Options

- A. 27.59%
- B. 16%
- C. 22%
- D. 36%

Correct answer: 27.59%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 27.59%.

Question 25 - Solved Example P32

Akhilesh has mean 21 and S.D. 5. His C.V. is approximately

Options

- A. 23.81%
- B. 5%
- C. 21%
- D. 27.59%

Correct answer: 23.81%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 23.81%.

Question 26 - Solved Example P32

Among Varad, Viraj and Akhilesh, the most consistent is

Options

- A. Varad
- B. Viraj
- C. Akhilesh
- D. all equally consistent

Correct answer: Varad

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives Varad.

Question 27 - Solved Example P32

If selection is based on the highest expected score, select

Options

- A. Viraj
- B. Varad
- C. Akhilesh
- D. none

Correct answer: Viraj

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives Viraj.

Question 28 - Solved Example P33

For company X shares with mean 50 and variance 7, S.D. is approximately

Options

- A. 2.64
- B. 7
- C. 5.28
- D. 1.90

Correct answer: 2.64

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 2.64.

Question 29 - Solved Example P33

For company X, C.V. is approximately

Options

- A. 5.28%
- B. 2.64%
- C. 7%
- D. 1.90%

Correct answer: 5.28%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 5.28%.

Question 30 - Solved Example P33

For company Y shares with mean 105 and variance 4, C.V. is approximately

Options

- A. 1.90%
- B. 4%
- C. 2.64%
- D. 5.28%

Correct answer: 1.90%

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives 1.90%.

Question 31 - Solved Example P33

The more stable company share is

Options

- A. company Y
- B. company X
- C. both equally stable
- D. neither

Correct answer: company Y

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives company Y.

Question 32 - Let's Construct P33

To compare word-length variability in a passage, calculate mean, S.D. and

Options

- A. coefficient of variation
- B. range only
- C. median only
- D. mode only

Correct answer: coefficient of variation

Explanation and steps

Use the stated definition or formula for dispersion. Substitution or direct identification gives coefficient of variation.